INCH/POUND

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SUPERSEDING
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# COMMERCIAL ITEM DESCRIPTION

# DRILL, RATCHET (REVERSIBLE)

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description covers hand-operated, reversible ratchet drills with American standard self-holding taper sockets.

2. CLASSIFICATION. Not applicable

3. SALIENT CHARACTERISTICS.

3.1 Design. The ratchet drills shall consist essentially of a handle and head, body, ratchet mechanism, feed screw, pawl, and feed pin. Ratchet drills shall be furnished with a short spindle with feed screw which enables ratchet to be converted into a boiler ratchet or adapter, to use square taper shank drills. Ratchet drills, when tested as specified, shall be capable of withstanding the applicable torque loads. Handles and tapered sockets shall show no indication of permanent set, cracks, or flaws and the ratchet mechanism and ratchet reverse control pawl engaging surfaces shall show no damage.

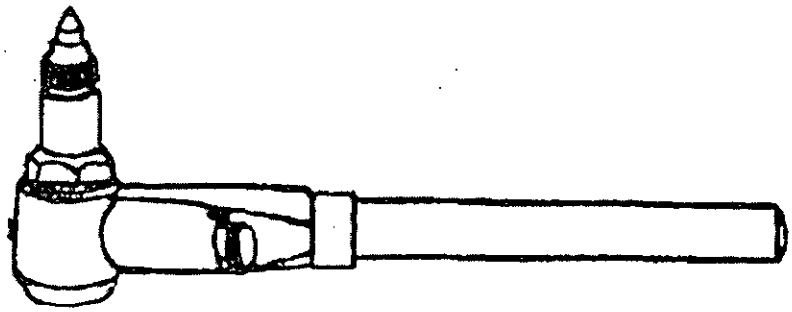


Figure 1. Reversible ratchet drill, American Standard self-holding taper socket

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Hardware SuperStore (6FEE), 1500 Bannister Rd. Bldg. #6, Kansas City, Missouri. 64131.

FSC 5110

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3.2 Handle and head. The handle and head shall be either integral, threaded or the head shall be welded securely to the handle. The handle shall be either a straight, round steel bar, or steel tubing smoothly finished to afford a comfortable handgrip. The head shall be drop-forged steel or malleable iron and all surfaces that bear or mate with the body shall have a satin finish.

3.3 Body. The body shall be drop-forged steel with a minimum of ten (10) teeth or notches on the outside diameter to assure ratcheting with the handle having a swing of 36 degrees or less. The inside tapered hole shall be either American Standard taper or bitstock taper to accommodate drill shanks within the capacity of the drill. The tapered hole shall be smoothly finished and shall be concentric with the bearing surfaces on the outside diameter. The outside diameter of the spindle or the end opposite the tapered hole shall be threaded to accommodate a body nut, or other means provided for securely holding the body in the head. The body shall form a part of the ratchet mechanism. The taper square hole shall be smooth and concentric with the bearing surface on the outside diameter.

3.4 Sockets. Sockets for ratchet drills shall have a standard self-holding taper in accordance with ASME B5.10.

3.5 Ratchet Mechanism. The ratchet mechanism shall be enclosed in the head and secured in such a manner as to minimize loosening during extreme usage. The ratchet mechanism shall be reversible by movement of a shifting lever, knurled button, or equal means which permits ratcheting operation of the body in either direction of rotation. The shifting lever or button shall be steel and shall be located in the head or at the handgrip end of the handle.

3.6 Feed Screw. The feed screw shall be alloy steel. The outside diameter of the screw shall be threaded with a class 3 thread for engaging the spindle end of the body. One end of the screw shall have a knurled collar with at least four (4) holes equally spaced at right angles to the axis to provide means for applying pressure to the drill. The knurled end of the screw shall be provided with an integral or removable hardened-steel center.

3.7 Pawl. The pawl shall be hardened alloy steel. The pawl shall be firmly attached to a rod or pin suitable for reversing the action of the ratchet. A spring or other means shall be provided for applying the proper tension on the pawl when engaged in the teeth or notches of the body.

3.8 Feed Pin. The feed pin shall be steel, suitable for regulating the feed screw in order to apply pressure to the drill.

3.9 Screw Threads. The screw threads used with the ratchet drills shall be in conformance with the screw-thread standards of FED-STD-H28.

3.10 Finish. All exposed metal surfaces of the ratchet drills shall have one of the coatings specified below. All external surfaces shall be free from pits, nodules, forge flash, cracks, burrs, and other detrimental defects.

3.10.1 Bright Chromium Plate. The coating shall be electro-deposited metals consisting of nickel followed by chromium. The minimum thickness shall be 0.0002 inch for nickel and 0.00001 inch for chromium.

3.10.2 Bright Zinc Plate. The coating shall be electro-deposited nickel. The minimum thickness shall be 0.0002 inch.



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3.10.3 Zinc Plate. The coating shall be electro-deposited zinc and shall be subjected to a chemical or electrochemical chromating. The minimum thickness shall be 0.0003 inch.

3.10.4 Hot Phosphate. The coating shall consist of chemically produced phosphate, followed by a coating of preservative.

3.10.5 Oxide Coating. The coating shall consist of chemically produced oxide, followed by a coating of preservative.

3.11 Identification Marking. Each drill shall be etched or stamped in permanent and legible manner with the manufacture's name or identifying symbol of such known character that the source may be readily identifiable, part number, and country of origin.

3.12 Workmanship. All drills shall be free from imperfections which may affect their safety, appearance, serviceability, or durability.

| NSN | Size Number | Height of head (max) inch | Length of feed (min) inch | Length of handle (min) inch | American Standard Taper Socket Number | Size of Drill inch | Torque Load (min) pound-inch |
|---|---|---|---|---|---|---|---|
| 5110-00-555-1784 | 1 | 5 | 2 | 9 | 1 | 1/16 to 9/16 | 2500 |
| 5110-00-223-5028 | 8 | 6-3/4 | 2-1/2 | 15 | 3 | 59/64 to 1-1/4 | 5500 |

Table I. American Standard self-holding taper socket

4. REGULATORY REQUIREMENTS.

4.1 Recovered Material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE.

5.1 Product Conformance. The product provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawing , specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Responsibility for Inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure the item conforms to the specified requirements.

6. PACKAGING. Preservation, packaging and marking shall be as specified in the contract or order.

7. NOTES. This section contains information of a general or explanatory nature that is not mandatory.

7.1 Source of Documents.



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American Society of Mechanical Engineers (ASME) Manuel Gutierrez, Managing Director, 3 Park Avenue, New York, NY 10016.

Federal standards are available from U.S. Government Printing Office, Washington, DC 20401-9371.

Federal Acquisition Regulation (FAR): Government Printing Office, Superintendent of Documents, Washington DC. 20401-9371.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents.

a. Title, number and date of Commercial Item Description.
b. National Stock Number and size number.
c. Packaging requirements.

MILITARY INTERESTS:
NONE: DoD has no registered interest in revisions and amendments to this Commercial Item Description until further notice.

Preparing Activity:
GSA-FSS



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